

DBMS

Queries in Relational Algebra

Branch (branch-name, branch-city, assets)

Customer (customer-name, customer-street, customer-city)

Account (account-number, branch-name, balance)

Loan (loan-number, branch-name, amount)

Depositor (customer-name, account-number)

Borrower (customer-name, loan-number)

Credit_info (customer-name, limit, credit_balance)

01. Find all loan made at “paryridge” branch.

Ans: $\sigma_{\text{branch_name} = \text{“paryridge”}}(\text{loan})$

02. Find all loan of over \$1200.

Ans: $\sigma_{\text{amount} > 12000}(\text{loan})$

03. Find all tuples who have taken loans of more than \$1200 made by the “paryridge” branch.

Ans: $\sigma_{\text{branch_name} = \text{“paryridge”} \wedge \text{amount} > 1200}(\text{loan})$

04. Find all loan numbers and amount of the loans.

Ans: $\pi_{\text{loan_number}, \text{amount}}(\text{loan})$

05. Find the loan number for each loan of an amount greater than \$1200.

Ans: $\pi_{\text{loan_number}}(\sigma_{\text{amount} > 1200}(\text{loan}))$

06. Find those customers who lives in “Harrison”.

Ans: $\pi_{\text{customer_name}}(\sigma_{\text{customer_city} = \text{“Harrison”}}(\text{Customer}))$

07. Find the names of all customers who have a loan, an accounts or both, from the bank.

Ans: $\pi_{\text{customer_name}}(\text{borrower}) \cup \pi_{\text{customer_name}}(\text{depositor})$

08. Find the names of all customers who have a loan and an account at bank.

Ans: $\pi_{\text{customer_name}}(\text{borrower}) \cap \pi_{\text{customer_name}}(\text{depositor})$

09. Find the names of all customers who have an account but no loan from the bank.

Ans: $\pi_{\text{customer_name}}(\text{borrower}) - \pi_{\text{customer_name}}(\text{depositor})$

10. Find the names of all customers who have a loan at the “perryridge” branch.

Ans: $\pi_{\text{customer_name}}(\sigma_{\text{branch_name} = \text{“perryridge”}}(\text{borrower} \bowtie \text{loan}))$

11. Find the names of all customers who have a loan at the bank. Along with the loan number and the loan amount.

Ans: $\pi_{\text{customer_name}, \text{loan.loan_number}, \text{amount}}(\text{borrower} \bowtie \text{loan})$

12. Find the names of all branches with customers who have an account in the bank and who live in Harrison.

Ans: $\pi_{\text{branch_name}}(\sigma_{\text{customer_city} = \text{“Harrison”}}(\text{Customer} \bowtie \text{borrower} \bowtie \text{loan}))$

13. Find all customers who have both a loan and an account at the bank.

Ans: $\pi_{\text{customer_name}}(\text{borrower} \bowtie \text{depositor})$

14. Find the names of all customer who have a loan at the “Perryridge” branch.

Ans: $\pi_{\text{customer_name}}(\sigma_{\text{branch_city} = \text{“Perryridge”}}(\text{borrower} \bowtie \text{loan}))$

15. Find all customers who have an account at all branches located “Brroklyn” city.

Ans: $r1 = \pi_{\text{branch_name}}(\sigma_{\text{branch_city} = \text{“Brroklyn”}}(\text{branch}))$

$r2 = \pi_{\text{customer_name}, \text{branch_name}}(\text{depositor} \bowtie \text{account})$

$r2/r1$.

16. Find how much each person can spend.

Ans: $\pi_{\text{customer_name, limit - credit_balance}}(\text{credit_info})$